DISTRIBUTED ALGORITHMS • FAULT-TOLERANT CONSENSUS • CRASH-STOP FAILURES

FloodSet Algorithm: Consensus Under Crash Faults

Synthesis based on Arpit Bhayani's Distributed Systems Masterclass • Lynch (Distributed Algorithms)

ALGORITHMIC PREMISE & THE CLEAN-ROUND GUARANTEE

Under the Crash-Stop failure model, servers can silently die at any instant—even halfway through broadcasting to peers. The FloodSet algorithm guarantees unanimous agreement among all non-faulty processes in a synchronous network tolerating up to f arbitrary node crashes. By executing for strictly **$f + 1$ rounds**, the Pigeonhole Principle guarantees that at least one round experiences *zero crashes* (the Clean Round), forcing all surviving processes to synchronize their knowledge sets.

1. THE CONSENSUS PROBLEM SPECIFICATION

Each node p_i starts with input $v_i \in \{0, 1\}$ and must decide on $d_i \in \{0, 1\}$ satisfying:

1. Agreement

All non-faulty processes output the exact same decision: $ orall p_i, p_j \implies d_i = d_j$.

2. Validity

If all nodes start with v, all decide v. Prevents trivial constant output.

3. Termination

All non-faulty nodes decide within bounded time ($f+1$ rounds).

2. WHY EXACTLY F + 1 ROUNDS? THE CLEAN-ROUND PRINCIPLE

Fischer and Lynch (1982) proved that any synchronous consensus protocol tolerating f crash failures requires at least $f + 1$ rounds in the worst case:

- An adversary has at most f crash budget.
- Across $f + 1$ rounds, by the Pigeonhole Principle, there must be at least one round with **zero crashes**.
- In this clean round, every message sent is delivered to all surviving nodes, synchronizing their states.

3. THE FLOODSET PROTOCOL MECHANICS

1. **State:** Each node p_i maintains a set $W_i \subseteq \{0, 1\}$, initialized to $W_i \gets \{v_i\}$.
2. **Rounds $r = 1 \dots f + 1$:** Node p_i broadcasts W_i to all nodes. Upon receiving sets M_j from peers, it takes the union: $W_i \gets W_i \cup igcup_j M_j$.
3. **Decision Rule:** At round $f + 1$:
   - If $|W_i| == 1$ ($W_i = \{v\}$): Decide v.
   - If $|W_i| == 2$ ($W_i = \{0, 1\}$): Decide default fallback 0.

4. FORMAL PROOF OF CORRECTNESS

1. Proof of Validity

Suppose all nodes start with $v \in \{0, 1\}$. Every initial set is $W_i = \{v\}$. Since no process ever sees $1-v$, no node ever adds $1-v$ to its set. At round $f+1$, every non-faulty node has $W_i = \{v\}$. Because $|W_i| = 1$, all nodes decide v. Holds.

2. Proof of Agreement via Clean Round

1. Let round $k \le f+1$ be the first round in which zero processes crash.
2. Every process alive in round k successfully transmits its set to all surviving nodes.
3. Therefore, all processes alive at the end of round k compute the exact same set union: $W_{ ext{clean}} = igcup_{p} W_p$.
4. In all subsequent rounds $k+1 \dots f+1$, nodes only broadcast subsets of $W_{ ext{clean}}$. No node can introduce any new value.
5. Thus, at round $f+1$, all non-faulty nodes hold **strictly identical sets**: $W_i = W_j = W_{ ext{clean}}$.
6. Because their sets are identical, they apply the exact same decision rule: if $|W_{ ext{clean}}| = 1$, both output v; if $|W_{ ext{clean}}| = 2$, both output fallback 0.

Agreement is mathematically guaranteed. ▪

5. FORMAL COMPLEXITY & BOUNDS

Time Complexity: Strictly $f + 1$ Rounds

By the Fischer-Lynch lower bound, consensus in f rounds is impossible. FloodSet achieves the optimal theoretical bound of exactly $f + 1$ rounds.

Message Complexity: $\mathcal{O}((f + 1) n^2)$

In each round, every alive process broadcasts its set to all n nodes ($\le n^2$ messages/round). Across $f + 1$ rounds: $\mathcal{O}((f + 1) n^2)$ messages.

6. PROTOCOL COMPARISON: SYNCHRONOUS VS. ASYNCHRONOUS CONSENSUS

Protocol	Network Model	Failure Model	Termination Guarantee	Round / Time Bound
FloodSet	Synchronous	Crash-Stop ($f < n$)	Deterministic	Strictly $f + 1$ rounds
Exponential Info Gathering (EIG)	Synchronous	Crash or Byzantine ($n \ge 3f + 1$)	Deterministic	Strictly $f + 1$ rounds
Paxos / Multi-Paxos	Asynchronous / Partial Sync	Crash-Recovery (Majority $f < n/2$)	Probabilistic under contention	Unbounded worst case; 1–2 RTTs normally
Raft	Partial Synchrony	Crash-Recovery (Majority $f < n/2$)	Randomized election timers	1 RTT during stable leader

7. SYSTEMS ENGINEERING TAKEAWAYS

1. **Crash Tolerant Round Horizons:** In synchronous networks, tolerating f crash faults fundamentally requires at least $f+1$ communication rounds to guarantee surviving a cascading mid-broadcast crash schedule.
2. **The Vulnerability of Flat Sets to Byzantine Faults:** FloodSet assumes honest crash-stop nodes. If a single node is Byzantine (maliciously lying about seeing $\{0, 1\}$), it corrupts the union set, forcing honest nodes into unwanted defaults.
3. **Foundation for Quorum Consensus:** The clean-round principle led directly to modern leader-lease and view-stamped replication architectures, where leader heartbeats ensure safe tenure during stable communication intervals.